P4 - Generative Store

ARC 3133-001 / Architectural Visualization Methods 1 / Fall 2026 - Full Term

Approach: Generative

Revision: 2026-11-02. Current assignment requirements and deadlines. Completed midterm work is reviewed using the evidence already submitted; no retrospective deliverables are added.

4.1 - The system - algorithm and diagram

Introduced: Week 11 - Nov 3 / **Due:** Final review - December 11, 2026 (no class in December) / **Weight:** 5%

Preparation checks: Week 12 - Nov 10: Approval in class: your generative strategy, your brand and a first test. Both strategy and brand must be approved on November 10.; Week 13 - Nov 17: Bring the system sheet in progress for review.

Purpose

Explain your generative system so that someone else could run it. The sheet is a diagram of the algorithm, not of a building.

What you will learn

- Define a generative system as a start, a rule and repetitions, and tell it apart from a procedure and from a parametric model.
- Explain a rule in plain words and in a diagram, step by step.
- Choose an environment and brand constraints that give the rule a reason to produce space.

What to do

- Pick one generative strategy from the overview (cellular automata, agents and stigmergy, reaction-diffusion, growth, physics solvers) or propose another. Pick a real brand for a store of 40 to 64 m2. Both are approved in class on November 10.
- Name the start, the rule and the repetitions. State what the rule reads (neighbours, a field, a trail, forces) and what the environment is (site, boundary, attractor, brand constraints).
- Draw the algorithm step by step, BIG-diagram style: one frame per step, in order, showing how the system produces spaces. Caption each frame in plain words; no code is required.
- Show two or three generations or iterations from the same run so the reader sees the rule acting on its own result.
- Keep the system editable and simple. Supplied groups and plugins are allowed; credit them and explain their inputs and outputs.

What to submit

- **Presentation:** one 17 x 11 inch landscape sheet, `Lastname_Firstname_P4a.pdf`, presented at the final review on December 11 and reviewed in progress on November 17.
- **The algorithm:** the start, the rule, the environment and the number of repetitions, each named and drawn. One frame per step, read in order, with a short caption in plain words.
- **Generations:** two or three states of the same run, side by side, with what changed between them marked.
- **Brand and program:** the brand, the store size (40 to 64 m2) and the constraints you gave the system.
- **Editable source:** the working system (.blend or .gh with its .3dm) with supplied groups and plugins credited, in `Lastname_Firstname_P4a/`.

What we assess

- A system another person could run from the sheet; a clear start, rule and environment; readable step-by-step diagrams.
- Course weight 5%. Digital rubric: understanding 30%, communication 30%, technical execution 25%, exploration 15%.

4.2 - The design space - matrix and selection

Introduced: Week 12 - Nov 10 / **Due:** Final review - December 11, 2026 (no class in December) / **Weight:** 5%

Preparation checks: Week 14 - Nov 24: Bring the design-space sheet in progress for review.

Purpose

Map part of the design space of your system, read it with architectural questions and select the result you will develop.

What you will learn

- Use repetition to run one system many times under controlled variation.
- Read generative results with architectural criteria rather than by taste alone.
- Select, vary and select again, and explain the choice.

What to do

- Build a matrix: one parameter changes across the columns, another down the rows (seed, threshold, rule, reach, steps). Use the same view, scale and display for every cell and label both axes with their values.
- Read the matrix with architectural criteria you name: floor area, room to move, headroom, entrance, a front to the street, light. Mark the results that fit a store of 40 to 64 m2.
- Select one result and show a second round of small variations of it (breed), then your final choice. Explain why in two or three sentences.
- Work already done on the October field matrix can be reused here when it comes from your system.

What to submit

- **Presentation:** one 17 x 11 inch landscape sheet, `Lastname_Firstname_P4b.pdf`, presented at the final review on December 11 and reviewed in progress on November 24.
- **Matrix:** at least 3 x 3, ideally 5 x 5, same view and scale in every cell, both parameters and their values labelled.
- **Readings:** the criteria you used, shown on the matrix (colour, marks or numbers), and the results that pass.
- **Selection:** the chosen parent, its variations and the final pick, with a short explanation.
- **Editable source:** the matrix definition that produces the cells, in `Lastname_Firstname_P4b/`.

What we assess

- A working matrix from your own system, controlled variation, readable comparison, named criteria and a reasoned selection.
- Course weight 5%. Digital rubric: understanding 30%, communication 30%, technical execution 25%, exploration 15%.

4.3 - The proposal - competition sheet

Introduced: Week 12 - Nov 10 / **Due:** Final review - December 11, 2026 (no class in December) / **Weight:** 20%

Preparation checks: Week 15 - Dec 1: Bring the competition sheet in progress for review.

Purpose

Present the selected result as a competition entry. This sheet carries 20%, four times each of the other two, and is judged on the quality of the drawings and the clarity of the story.

What you will learn

- Develop a generative result into an architectural proposal without losing what the system produced.
- Post-process drawings for a competition audience: linework, colour, hierarchy, annotation.
- Tell one story across several drawings on one sheet.

What to do

- Develop the selected result into a store of 40 to 64 m2 for your brand: entrance, a front, usable floor, headroom, light. Keep what the system produced legible; post-process, do not redraw.
- Draw at minimum one elevation, one section, one isometric and one perspective of the same model. Add a person, key dimensions, the brand and short annotations.
- Compose the sheet as a competition entry: one story, a clear hierarchy, balanced colour, consistent linework, in your Graphic Standards Manual.
- Reference: the winning sheets of the Museum of Emotions and the Last Nuclear Bomb Memorial competitions (see Resources) set the level of drawing quality and post-processing we are looking for.

What to submit

- **Presentation:** one 17 x 11 inch landscape sheet, `Lastname_Firstname_P4c.pdf`, presented at the final review on December 11 and reviewed in progress on December 1.
- **Drawings:** one elevation, one section, one isometric and one perspective at minimum, all from the same model, post-processed, with a person, dimensions and the brand.
- **Story:** a title, two or three sentences on the system and the selection, and annotations that connect the drawings to the rule.
- **Editable model:** the selected result as generated plus the developed model, in `Lastname_Firstname_P4c/`.

What we assess

- Competition-sheet rubric: linework, quality of drawings, balance of color, layout, hierarchy, annotation, clarity of story, clarity of diagrams, quality of overall presentation.
- Course weight 20%. Also assessed: fidelity to the generated result and a convincing store at 40 to 64 m2.

Final booklet - submission checklist

Due: Final review - December 11, 2026 (no class in December) / **Weight:** 15%

Reuse the assignment sheets and document selected revisions.

- Submit `Lastname_Firstname_BOOK.pdf` as a 17 x 11 inch landscape booklet. Include a cover, contents, all four projects in assignment order, references and brief captions identifying substantive revisions. There is no fixed total page count.
- Reuse the assignment sheets: 1 GSM; 2.1 process and 2.2 print documentation; 3.1 module; 3.2.1-3.2.3; 3.4 studies and 3.5 application; 4.1 the system, 4.2 the design space and 4.3 the proposal. Include each shared sheet once. Completed midterm evidence remains valid.
- Include an overall image and useful detail of the CSG print, with short process/testing notes. Keep labels, typography and references consistent. The booklet is read with the competition-sheet rubric (linework, quality of drawings, balance of color, layout, hierarchy, annotation, clarity of story, clarity of diagrams, overall presentation).
- Submit `Lastname_Firstname_GSM.pdf` separately with the revised four-page manual, plus the editable booklet layout and linked assets. Organize native project models in their assignment folders inside `Lastname_Firstname_BOOK/`.
- Open the PDF and source from the final package; check page order, missing links, readable images and drawing scales. Present the booklet, the revised GSM and the three Project 4 sheets at the final review on December 11; there is no class in December.

Shared standards: Apply the GSM and credit sources. Package linked assets; check PDFs and editable files before uploading. Explain rules plainly and follow the course submission policies.